SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING) Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1005	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Date:	14-10-2026	Duration:	60 minutes

Code of Conduct

I _K.FUZAAIL - 192425159_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an Agile Sprint Planning and User StoryWorkshop by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an Agile Sprint Planning and User Story Workshop by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. Identify the Project Vision and Stakeholders
    - Define the project's vision, objectives, and identify all key stakeholders involved.
2. Create Epics and User Stories
    - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>*.
3. Define Acceptance Criteria
    - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. Prioritize the Product Backlog
    - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. Plan the Sprint Backlog
    - Select high-priority user stories for the first sprint and divide them into actionable development tasks. 6. Estimate Effort Using Story Points
    - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.

7. Present the Sprint Goal and Expected Deliverables
    - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified,sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation orsprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision &Stakeholder Identification (30%)	/5
Epics & User Stories (25%)	/5
AcceptanceCriteria (20%)	/5
Product Backlog Prioritization & Sprint Planning (15%)	/5
Story Point Estimation, Sprint Goal & Presentation (10%)	/5
TOTAL	/25

INTELLIGENT FLEET DRIVER SAFETY MANAGEMENT SYSTEM

Agile Sprint Planning and User Story Workshop

1. Project Vision and Stakeholder Identification

1.1 Project Vision

The vision of the Intelligent Fleet Driver Safety Management System is to develop a centralized platform for monitoring driver behavior, identifying unsafe driving practices, and improving fleet safety. The system will track driving activities such as speeding, driver fatigue, and harsh braking to identify potential risks.

The platform aims to help fleet managers monitor driver performance, generate safety scores, schedule driver training, and prepare safety reports. It will reduce accidents, improve driver accountability, and support safer fleet operations.

1.2 Project Objectives

- Monitor driver behavior and driving activities.

- Detect speeding, fatigue, and harsh braking.
- Generate driver safety scores.
- Identify high-risk drivers and driving patterns.
- Provide alerts for unsafe driving behavior.
- Schedule training for drivers who require improvement.
- Generate fleet safety and performance reports.

1.3 Stakeholder Identification

Stakeholder	Role
Drivers	Follow safety guidelines and review driving performance.
Fleet Managers	Monitor drivers, manage schedules, and review safety scores.
Safety Officers	Investigate safety incidents and recommend corrective actions.
System Administrator	Manage users, permissions, and system settings.
Organizational Management	Review reports and make fleet safety decisions.

2. Epics and User Stories

The project requirements are divided into five epics representing the major functions of the system.

Epic 1 – Driver and User Management

ID	User Story
US01	As a fleet manager, I want to securely log in so that I can access driver safety information.
US02	As an administrator, I want to manage driver profiles so that driver information remains accurate.

Epic 2 – Driver Behavior Monitoring

ID	User Story
US03	As a fleet manager, I want to monitor driver activities so that unsafe driving behavior can be identified.

US04	As a safety officer, I want to detect speeding, fatigue, and harsh braking so that potential risks can be addressed.
US05	As a fleet manager, I want to receive alerts for unsafe driving so that corrective action can be taken quickly.

Epic 3 – Safety Score and Risk Assessment

ID	User Story
US06	As a fleet manager, I want to generate driver safety scores so that driver performance can be evaluated.
US07	As a safety officer, I want to identify high-risk drivers so that accident risks can be reduced.

Epic 4 – Driver Training and Scheduling

ID	User Story
US08	As a fleet manager, I want to schedule driver safety training so that drivers can improve their behavior.
US09	As a driver, I want to view my training schedule so that I can attend assigned sessions.

Epic 5 – Analytics and Reporting

ID	User Story
US10	As a fleet manager, I want to view fleet safety statistics so that I can monitor overall performance.
US11	As organizational management, I want to generate safety reports so that I can make informed decisions.

Epic Summary

Epic	Main Purpose
Epic 1	Driver profiles and secure access
Epic 2	Monitoring and detecting unsafe driving
Epic 3	Safety scores and risk assessment

Epic 4	Driver training and scheduling
Epic 5	Safety analytics and reporting

3. Acceptance Criteria

Acceptance criteria define the conditions that must be satisfied for a user story to be completed. The Given–When–Then format is used.

US01 – Secure Login

- Given a user has valid credentials,
- When the user enters the login details,
- Then the system should authenticate the user and display the appropriate dashboard.

US02 – Driver Profile Management

- Given an administrator is logged in,
- When the administrator adds or updates driver details,
- Then the system should save the information correctly.

US03 – Driver Monitoring

- Given driver activity data is available,
- When the fleet manager opens the monitoring dashboard,
- Then the system should display the available driving information.

US04 – Unsafe Driving Detection

- Given driving data is available,
- When speeding, fatigue, or harsh braking is detected by the monitoring system,
- Then the system should record the event and identify the safety risk.

US05 – Safety Alerts

- Given an unsafe driving event meets the configured alert conditions,
- When the event is detected,
- Then the system should notify the appropriate fleet manager or safety officer.

US06 – Driver Safety Score

- Given sufficient driving data is available,
- When the system evaluates driver performance,
- Then it should generate a safety score based on defined safety criteria.

US07 – High-Risk Driver Identification

- Given driver safety scores and incident records are available,
- When the safety officer reviews driver performance,
- Then the system should identify drivers who meet the configured high-risk criteria.

US08 – Training Scheduling

- Given a driver requires safety training,
- When the fleet manager schedules a session,
- Then the system should save the schedule and associate it with the driver.

US09 – View Training Schedule

- Given a training session has been assigned,
- When the driver views the schedule,
- Then the system should display the session date and details.

US10 – Fleet Safety Statistics

- Given fleet safety records are available,
- When the fleet manager opens the dashboard,
- Then the system should display relevant statistics and safety scores.

US11 – Safety Reports

- Given sufficient historical safety data is available,
- When management requests a report,
- Then the system should generate a report containing relevant safety incidents and performance information.

Acceptance Criteria Summary

ID	User Story	Main Acceptance Result
US01	Secure Login	User authenticated
US02	Driver Profiles	Driver details saved
US03	Driver Monitoring	Driving data displayed
US04	Unsafe Driving Detection	Safety events recorded
US05	Safety Alerts	Appropriate alerts generated
US06	Safety Score	Driver score generated
US07	Risk Assessment	High-risk drivers identified

US08	Training Schedule	Training session saved
US09	View Schedule	Training details displayed
US10	Safety Statistics	Dashboard statistics displayed
US11	Safety Reports	Safety report generated

4. Product Backlog Prioritization

The product backlog is prioritized according to business value, safety importance, dependencies, and implementation effort. Fibonacci story points are used to estimate the relative effort.

4.1 Prioritized Product Backlog

Priority	ID	User Story	Priority Level	Story Points
1	US01	Secure user login	Must Have	3
2	US02	Manage driver profiles	Must Have	3
3	US03	Monitor driver activities	Must Have	5
4	US04	Detect unsafe driving	Must Have	5
5	US05	Generate safety alerts	Must Have	3
6	US06	Generate driver safety scores	Must Have	5
7	US07	Identify high-risk drivers	Should Have	5
8	US08	Schedule driver training	Should Have	3
9	US09	View training schedule	Should Have	2
10	US10	View fleet safety statistics	Should Have	5
11	US11	Generate safety reports	Could Have	5

4.2 Prioritization Logic

The development sequence is:

Login → Driver Profiles → Driver Monitoring → Unsafe Driving Detection → Safety Alerts → Safety Scores → Risk Assessment → Training → Analytics and Reports

Secure access and driver profiles establish the foundation. Monitoring and unsafe-driving detection provide the core safety functionality. Safety scores, training, and reporting can be developed after the essential monitoring functions are available.

4.3 Sprint 1 Selection

Sprint 1 focuses on the basic foundation of the system.

Selected Story	Story Points
US01 – Secure Login	3
US02 – Manage Driver Profiles	3
US03 – Monitor Driver Activities	5
Total	**11**

5. Sprint Backlog Planning

5.1 Sprint Details

- **Sprint:** Sprint 1
- **Duration:** 2 weeks
- **Sprint Goal:** Develop the basic system for secure access, driver profile management, and driver activity monitoring.
- **Sprint Capacity:** 11 story points

5.2 Task Breakdown

US01 – Secure Login

1. Design the login interface.
2. Create the user database.
3. Implement username and password authentication.
4. Add login validation and logout.
5. Test valid and invalid login scenarios.

US02 – Manage Driver Profiles

1. Design the driver profile interface.
2. Create the driver database structure.

3. Add and update driver information.
4. Validate required driver details.
5. Test driver profile creation and updating.

US03 – Monitor Driver Activities

1. Design the driver monitoring dashboard.
2. Define the required driving activity data fields.
3. Implement storage and retrieval of driver activity records.
4. Display available driver activity information.
5. Test the monitoring dashboard using sample data.

5.3 Sprint Backlog Summary

ID	User Story	Main Tasks	Points	Status
US01	Secure Login	Interface, authentication, validation, testing	3	To Do
US02	Driver Profiles	Database, profile management, validation, testing	3	To Do
US03	Driver Monitoring	Dashboard, activity records, data display, testing	5	To Do
Total			**11**	

6. Story Point Estimation

The Fibonacci sequence (1, 2, 3, 5, 8, 13) is used for relative effort estimation. Story points consider complexity, effort, and uncertainty.

6.1 Estimation Table

ID	User Story	Points	Justification
US01	Secure Login	3	Requires authentication, validation, and testing.
US02	Driver Profiles	3	Requires database setup and profile management functions.
US03	Driver Monitoring	5	Requires activity data handling, dashboard development, and testing.
Total		**11**	

6.2 Estimation Justification

- **US01 – 3 points:** Login functionality requires a basic interface, authentication, and validation.
- **US02 – 3 points:** Driver profile management requires storing, retrieving, and updating driver information.
- **US03 – 5 points:** Driver monitoring requires handling activity data and displaying it clearly on the dashboard.

Planning Poker can be used by the team to discuss and agree on the story-point estimates.

Sprint 1 Capacity: 11 Story Points

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

"To develop a basic Intelligent Fleet Driver Safety Management System that supports secure login, driver profile management, and monitoring of available driver activity data."

7.2 Expected Deliverables

At the end of Sprint 1, the expected deliverables are:

1. **Login Module:** Secure access for authorized users.
2. **Driver Profile Module:** Ability to add and update driver information.
3. **Driver Database:** Storage for driver profiles.
4. **Monitoring Dashboard:** Display of available driver activity records.
5. **Data Validation:** Basic validation of user and driver information.
6. **Testing:** Functional testing of login, profiles, and monitoring.
7. **Working Prototype:** Integration of the completed Sprint 1 features.

Note: Actual fatigue detection, speeding detection, and harsh-braking alerts depend on suitable driving data and, where required, vehicle sensors or telematics integrations. Sprint 1 establishes the foundation for these advanced features.

7.3 Sprint 1 Workflow

User Login → Manage Driver Profiles → View Driver Activity Dashboard

7.4 Definition of Done

A user story is considered complete when:

- The required functionality is implemented.
- Data is stored and retrieved correctly.
- Basic validation is completed.

- Functional testing is successful.
- No major defects remain.
- The feature works with the Sprint 1 prototype.

7.5 Sprint 1 Summary

Item	Details
Sprint	Sprint 1
Duration	2 weeks
Goal	Build the basic fleet safety management foundation
Selected Stories	US01, US02, US03
Total Story Points	11
Main Deliverables	Login, driver profiles, monitoring dashboard
Expected Outcome	Working basic fleet safety management prototype

Conclusion

The Intelligent Fleet Driver Safety Management System aims to improve fleet safety through driver monitoring, unsafe-driving detection, safety scoring, training, and reporting. Agile sprint planning divides the project into manageable user stories and prioritizes the most important functions. Sprint 1 establishes the basic foundation, enabling future sprints to implement advanced safety monitoring, alerts, risk assessment, and analytics.